

Modelling Pitch Expectation in Melody: A Comparison of Variable-Order Markov and Transformer-Based Approaches

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Introduction

- Expectation is widely regarded as a key mechanism underlying the emotional impact of music.
- Some models of expectation capture **statistical learning** (the identification of statistical regularities from a lifetime of musical listening, and from the piece at hand).
- We compare a **probabilistic model** (IDyOM) with a **deep-neural** approach (Music Transformer) to assess how well they capture pitch expectation in melody.

Models

- IDyOM**: estimates the conditional probability of each possible next note given the preceding musical context, using n-gram modelling.
- Trained on corpus of monophonic melodies.
- Different IDyOM models can be created using various **viewpoints** (enables the tracking of various pitch and rhythm characteristics).
- Music Transformer**: originally purposed for generative music.
- Decoder-only** architecture, uses **relative self-attention** to predict the next musical event from a sequence of symbolic tokens.
- Trained on MAESTRO and fine-tuned on MCCC.

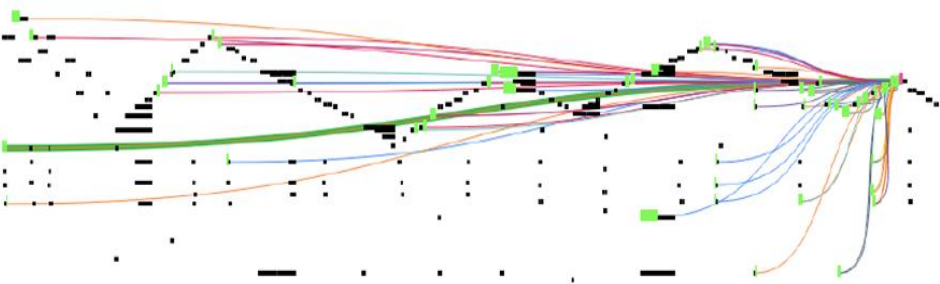


FIG 1. A piano-roll visualisation showing which musical events most strongly affect a prediction made by the Music Transformer

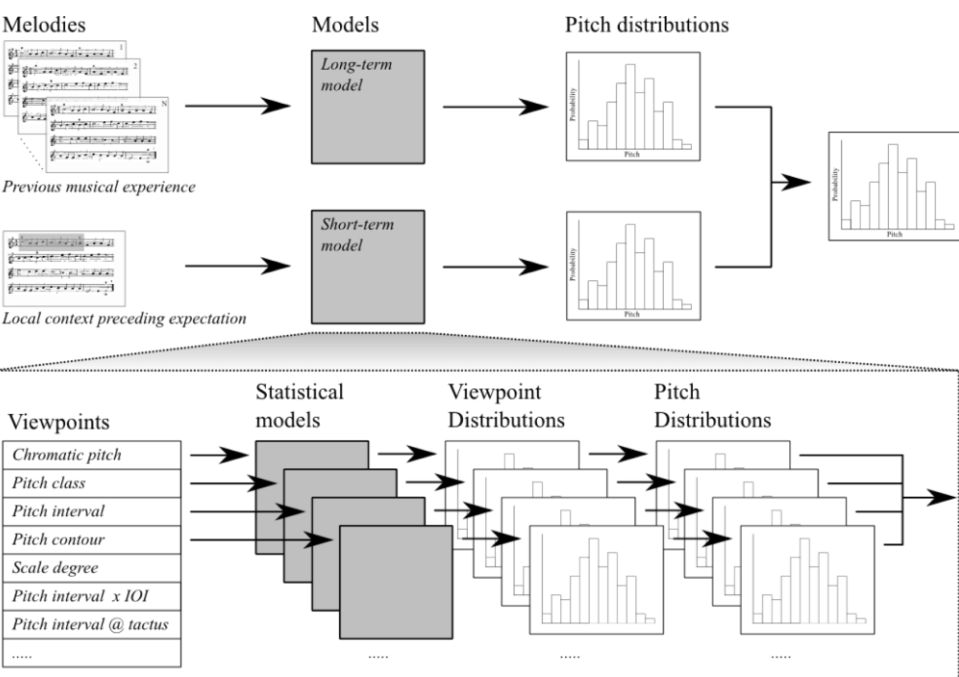


FIG 2. A visualisation of how IDyOM combines various viewpoints in a long-term and short-term model to generate predictions

Methodology

- OLS regression** to quantify how closely the models' predictions aligned with human ratings of expectation.
- Probe-tone ratings for two-note and longer melodic stimuli**; continuous ratings on Bach chorale melodies.
- Unique variance partitioning** based on two-predictor regression.
- Comparison of correlations** (Steiger's test).

Model	Viewpoints
IDyOM 1	(cpitch)
IDyOM 2	((cpint dur) cpintfip cpcint)
IDyOM 3	((cpint cpintfref))
IDyOM 4	(cpintfib cpintfip (cpint cpintfref) (cpitch ioi))
IDyOM 5	(cpintfip (cpintfref dur-ratio) thr-cpint-fiph)

FIG 2. Viewpoints chosen in each IDyOM model used

Results

- Two-note stimuli**: the Music Transformer significantly outperformed IDyOM 1; the optimised IDyOM 2 significantly outperformed the Transformer.
- Longer stimuli**: IDyOM 4 (optimised) slightly outperformed the Transformer, but the difference was not statistically significant.
- Mostly overlapping variance** in all cases.
- Continuous ratings**: IDyOM significantly outperformed Transformer; when excluding first 7 notes no significant difference.

Stimulus	IDyOM model	R^2 (IDyOM)	R^2 (Transformer)	R^2 (both)	Unique IDyOM	Unique Transformer	Z	p
Two-note	1	0.367	0.559	0.581	0.022	0.214	3.5796	0.003
Two-note	2	0.687	0.559	0.701	0.142	0.014	-3.3742	0.0007
Long	1	0.679	0.664	0.782	0.118	0.103	0.2685	0.7883
Long	3	0.554	0.664	0.714	0.050	0.160	-1.8755	0.0607
Long	4	0.729	0.664	0.786	0.122	0.057	1.3385	0.1807
Continuous	5	0.424	0.059	0.424	0.365	0	-2.6825	0.0073
Continuous -7	5	0.353	0.231	0.408	0.177	0.055	-0.8605	0.055

TABLE 1. Results table showing the coefficient of determination (R^2) for each regression, the unique contributions of each model, and the Z and p values from the Steiger's test

Conclusions

- The **Music Transformer modelled melodic expectation competently**, but without outperforming the optimised IDyOM models; worse performance for early notes in melodies.
- The low unique variances suggest that the two models tend to predict the same context-probe pairs well across many stimuli.
- Could indicate that they use similar representations, but further work is needed to render the Transformer's predictions more interpretable.
- Such speculation can be guided by examining which notes in the context receive higher softmax probability.